THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Some Applications of Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $6\sqrt{3}$ m above the level of the person's eyes and the eyes are 18 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 45° **(b)** $30\sqrt{3}^\circ$ **(c)** 60° **(d)** 30°

2. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 14 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. The height of the tower is: 1

(a) $7 + 7\sqrt{3}$ m **(b)** 14 m **(c)** $\frac{14\sqrt{3}}{3}$ m **(d)** $21 + 7\sqrt{3}$ m

3. A flagstaff stands on the top of a building 24 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) $48\sqrt{3}$ m **(b)** 96 m **(c)** 72 m **(d)** 48 m

4. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 16 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) 8 minutes **(b)** $\frac{16}{3}$ minutes **(c)** 16 minutes **(d)** 32 minutes

5. From the top of a tower $15\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) 60 m **(b)** 15 m **(c)** $30\sqrt{3}$ m **(d)** 30 m

6. From the top of a building 21 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. The height of the tower is: 1

(a) $7\sqrt{3}$ m **(b)** 7 m **(c)** 28 m **(d)** $28\sqrt{3}$ m

7. Two points on level ground lie on opposite sides of a tower of height 27 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. The distance between the two points is: 1

(a) $27 - 9\sqrt{3}$ m **(b)** 36 m **(c)** $27 + 9\sqrt{3}$ m **(d)** $27 + 27\sqrt{3}$ m

8. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $7\sqrt{3}$ m above the level of the person's eyes and the eyes are 21 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) $30\sqrt{3}^\circ$ **(b)** 30° **(c)** 60° **(d)** 45°

9. **Assertion (A):** A 10 m ladder leaning against a wall makes an angle of 60° with the ground. It reaches a height of $5\sqrt{3}$ m. 1

Reason (R): In a right triangle, $\sin\theta = \dfrac{\text{perpendicular}}{\text{hypotenuse}}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A kite string 100 m long, taut and inclined at 30° to the ground, has the kite at a height of 50 m. 1

Reason (R): $\sin 30^\circ = \frac{\sqrt{3}}{2}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 14 minutes. Find the further time the car takes to reach the foot of the tower. 2

12. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 22 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 2

13. A flagstaff stands on the top of a building 27 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 30° respectively. Find the length of the flagstaff. 2

14. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 22 minutes. Find the further time the car takes to reach the foot of the tower. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A building is $26\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

16. A flagstaff stands on the top of a building 24 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

17. Two points on level ground lie on opposite sides of a tower of height 36 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. Find the distance between the two points. 3

OR

A building is $28\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

18. A flagstaff stands on the top of a building 30 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

OR

Two points on level ground lie on opposite sides of a tower of height 42 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. Find the distance between the two points. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** From the top of a 60 m high lighthouse, the angles of depression of two ships on the same side of the lighthouse (in line with its foot) are 60° and 30°. (Use $\sqrt{3} = 1.73$.) 5

(i) What is the angle of elevation of the top of the lighthouse from the nearer ship?
(ii) Find the distance of the nearer ship from the foot of the lighthouse.
(iii) Find the distance of the farther ship from the foot of the lighthouse.
(iv) Find the distance between the two ships.

20. A building is 14 m high. From the top of the building the angle of elevation of the top of a tower is 45° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 5

OR

Two vertical poles of equal height stand on opposite sides of a road 192 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

END OF QUESTION PAPER — Total: 40 Marks